

Part 4: NLP Interview Cheatsheet (8 Essential Must-Know Foundations)

A high-density revision reference for AI, Applied AI, GenAI, and ML Engineer interviews covering the 8 essential formulas, core diagrams, time & memory complexity, interview Q&A, and a 1-page comparison table.

1. The 8 Must-Know Essential Formulas

#	Topic	Must-Know Mathematical Formula
1	TF-IDF Vectorization	$TF(t, d) = \frac{\text{count}(t, d)}{ d }, \quad IDF(t) = \log\left(\frac{1+ D }{1+DF(t)}\right) + 1, \quad \text{TF-IDF} = TF \times IDF$
2	Cosine Similarity	$\text{CosineSim}(u, v) = \frac{u \cdot v}{\ u\ _2 \ v\ _2}$
3	Naive Bayes (+ Laplace)	$P(c \mid d) \propto P(c) \prod P(w_i \mid c), \quad P(w_i \mid c) = \frac{N_{c,i} + 1}{N_c + V }$
4	Word2Vec (Skip-Gram)	$P(w_O \mid w_I) = \sigma(v'_{w_O}{}^\top v_{w_I}) = \frac{1}{1 + \exp(-v'_{w_O}{}^\top v_{w_I})}$
5	LSTM Cell Updates	$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f), \quad i_t = \sigma(W_i[h_{t-1}, x_t] + b_i), \quad C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t, \quad h_t = o_t \odot \tanh(C_t)$
6	Scaled Dot Attention	$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V$
7	BLEU-2 Score	$\text{BLEU-2} = \text{BP} \cdot \exp(0.5 \log p_1 + 0.5 \log p_2), \quad \text{BP} = \exp(1 - r/c) \text{ if } c \leq r \text{ else } 1.0$
8	Perplexity (PPL)	$\text{PPL} = \exp(\mathbb{E}[-\log P(w_i)]) = \left(\prod P(w_i \mid w_{-i})\right)^{-1}$

2. Time & Memory Complexity Matrix

Algorithm / Architecture	Training Time Complexity	Inference Time Complexity	Memory Footprint Complexity
TF-IDF Indexing	$O(D \times L)$	$O(k \times V)$	$O(D \times V)$ Sparse
Multinomial Naive Bayes	$O(D \times L)$	$O(C \times L)$	$O(C \times V)$ Dense
Word2Vec Skip-Gram	$O(C \cdot m \cdot K \cdot d)$	$O(1)$ Lookup	$O(V \times d)$ Embedding Table
LSTM Model Layer	$O(T \cdot d^2)$ per sequence	$O(T \cdot d^2)$ Sequential	$O(4 \cdot d^2)$ Model Weights
Scaled Dot-Product Attention	$O(T^2 \cdot d)$ per batch	$O(T^2 \cdot d)$	$O(T^2)$ RAM Attention Weights
INT8 Quantized Serving	$O(N \cdot d)$ SIMD	$O(N \cdot d)$ SIMD	$O(\frac{1}{4}M_{\text{FP32}})$ Bytes

3. High-Frequency Interview Q&A Summary

1. Why do we scale $QK^\top$ by $\sqrt{d_k}$ in Attention?

- *Answer:* Large d_k causes dot products $q \cdot k$ to have variance d_k . Large dot products push Softmax into saturated regions with near-zero gradients ($\text{softmax}'(z) \rightarrow 0$), causing vanishing gradients during backpropagation. Dividing by $\sqrt{d_k}$ rescales variance back to 1.0.

2. Why does Laplace Smoothing prevent zero probability in Naive Bayes?

- *Answer:* Unseen vocabulary words in a training class yield raw likelihood $P(w_{\text{new}} | c) = 0$, causing the entire posterior product to collapse to zero. Adding +1 ensures every word retains a positive non-zero likelihood $P(w_i | c) = \frac{N_{c,i}+1}{N_c+|V|}$.

3. How does the LSTM cell state C_t resolve vanishing gradients?

- *Answer:* Cell update $C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t$ is additive. When forget gate $f_t = 1$, the gradient derivative $\frac{\partial C_t}{\partial C_{t-1}} = 1$, preserving gradient norm ≈ 1.0 across long sequences.

4. Why is lower Perplexity better?

- *Answer:* $\text{PPL} = \exp(\mathcal{L}_{\text{CE}})$. Perplexity represents the average branching factor of uncertainty when predicting the next token. A lower PPL means fewer candidate choices the model is uncertain between.

5. Why is BLEU precision-oriented while ROUGE is recall-oriented?

- *Answer:* BLEU checks what fraction of generated candidate n-grams are valid (preventing hallucinated ungrounded text in translation). ROUGE checks what fraction of reference target tokens were captured in the summary (preventing key detail omission).